D02 · Demand Planning & Forecasting

03 · Promotional & event-driven forecasting

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Promotional forecasting separates baseline demand from event lift — cleansing history of past promotions, modeling uplift per mechanic (price reduction, display, feature), and accounting for cannibalization and the post-promo dip — instead of layering a flat percentage on a contaminated base.

2 WHY IT MATTERS

Fildes & Goodwin **(Fildes & Goodwin, 2007 — peer-reviewed)** showed that judgmental promotional adjustments are systematically optimistic and almost never tracked, so error compounds unchecked. If past promo spikes are left in the baseline, the statistical model reads them as level and over-forecasts the non-promo weeks; the new promo lift is then stacked on top, double-counting the spike. Add the ignored post-promo trough — pantry loading borrows demand from future weeks — and substitute cannibalization, and promo-period error of **40–60%** is routine, landing as both promo stockouts (lost lift) and post-promo markdown (lost margin).

3 HOW IT IS DONE

1. **Cleanse the baseline first.** Flag every historical promo period, remove the spike, interpolate the underlying baseline, and store the promotions as a separate event series. A baseline contaminated by old promos poisons every forecast built on it.
2. **Model lift per mechanic, not as one number.** A temporary price reduction, an end-cap display, and a feature ad have different multipliers and different decay. Estimate each from history; a single blended uplift averages away the levers you actually pull.
3. **Model cannibalization explicitly.** When SKU A goes on promo, substitutes B and C dip. Forecast the category, then split — promoting A in isolation over-forecasts A and starves the replenishment of B and C.
4. **Model the post-promo dip.** Pantry loading pulls demand forward; the weeks after a promo run below baseline. Size the trough or you over-replenish straight into it and create the next markdown.
5. **Lock the promo calendar into the forecast on a fixed cadence.** Late promo entry is the single largest source of promo error — supply cannot react to a promotion it learns about two weeks out. Freeze the calendar at the planning horizon.
6. **Track promo FVA.** Compare the promoted forecast against the cleansed baseline forecast and against actuals. Promotions whose forecast does not beat baseline accuracy are being managed by hope; FVA exposes them.

> **WORKED EXAMPLE — ILLUSTRATIVE**
>
> A grocery retailer forecast promos as **+35% flat on a non-cleansed baseline** — **baseline: promo-week MAPE 52%, stockouts on 18% of promo events, $1.4M/quarter post-promo markdown**. It cleansed history, modeled lift per mechanic, and added cannibalization and dip. Results: **promo-week MAPE 52% -> 29% (-44%), promo stockouts 18% -> 9% of events, markdown $1.4M -> $0.9M/qtr (-36%)** — the cleansed baseline alone removed half the error before any promo modeling.

IN PRACTICE

Promo calendar lock cadence

Agree a hard cut-off after which the promo calendar is frozen into the forecast — typically one full supply lead time out. Promotions added inside that window are run at risk, on existing stock, with no forecast adjustment. Late entry is the dominant cause of promo stockout.

Category-then-split forecasting

Forecast total category demand for the promo period, then split across SKUs. This captures cannibalization automatically — the promoted SKU gains share within a roughly conserved category total — instead of treating each SKU as if it lived alone.

Promo FVA tracking

Maintain a post-event scorecard: planned lift vs actual lift, by mechanic and by category. Over time this becomes the uplift library and the evidence base for which promotions are worth running — and which are repeatedly over-forecast.

Uplift libraries by mechanic and seasonality

Build and maintain a structured library of historical lifts indexed by mechanic, depth, category, and season. A new promotion is forecast by matching it to the library, not by a planner's memory of the last one — turning institutional knowledge into a reusable model.

WHAT YOU WOULD USE

PROMOTIONAL / TRADE-PROMO FORECASTING

SAP IBP + TPM / o9 — enterprise tier-1 — Promotional demand modeling integrated with trade-promotion management and the full S&OP plan, with baseline cleansing and lift decomposition. **(SAP, o9 — vendor)** The choice when promo planning must reconcile with financial and supply plans.

RELEX / Blue Yonder — strong in grocery/retail — Retail-native promotional forecasting with mature baseline cleansing, cannibalization, and store-level lift modeling. **(RELEX, Blue Yonder — vendor)** The fit when promotions are frequent, store-specific, and the replenishment loop is tight.

Trade-promotion optimization specialists — Dedicated TPO engines that model promotion ROI and lift at depth. **(specialized TPO vendors — vendor)** The vertical option for CPG manufacturers where trade spend is a top-three P&L line.

For operations that don't need a platform — A cleansed-baseline workbook plus a lift-multiplier table indexed by mechanic, with a manual cannibalization and dip adjustment. The decomposition discipline — separate base, separate lift — matters far more than the tool that holds it.

4 SO WHAT

Operations that decompose baseline from lift price the promotion before they run it; operations that stack a flat percentage on a dirty baseline pay twice — stockout during the lift, markdown after it. The flat-percentage uplift is the most common promo forecast and the most expensive — it ignores the mechanic, the cannibalization, and the trough, then quietly poisons next quarter's baseline. A promotion you cannot forecast cleanly is a promotion you are running on faith.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México